

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION

Question Paper

Month and Year of Examination: August 2026

Programme: B.Tech

Semester: 4th

Subject: **Strength of Materials**

Course No.: PIPC211

Time Allotted: 3 Hrs.

Maximum Marks: 50

Assume suitably and state, additional data required, if any.

Attempt any five questions. Each question carries 10 marks.

- 1. Two steel plugs fit freely into the ends of a steel tubular distance piece 400 mm long and are drawn together by a steel bolt 500 mm long and nut, the nut being tight fit in the beginning. The nut is further tightened by a quarter of a turn to draw the pieces together, the pitch of the bolt thread being 2 mm. The pieces are then subjected to forces of 50 kN tending to pull them apart. Calculate the stresses in the bolt and the tube. The area of cross-section of the bolt is 700 sq. mm and that of the tube is 500 sq. mm. $E_s = 200 \text{ GN/m}^2$.
- 2. At a point in a material under stress, the intensity of the resultant stress on a certain plane is 100 MN/m^2 (compressive) inclined at 40° to the normal of that plane. The stress on a plane at right angles to this has a normal tensile component of intensity of 60 MN/m^2 . Find
 - a. Find normal and shear stresses on planes inclined at 25° and 65° in the clockwise direction to 100 MN/m^2 stress.
 - b. The principal stresses and their direction.
 - c. The shear stresses and their direction along with the magnitude of normal stress on the plane of maximum shear stress.

Show the results on suitably oriented elements and state the sign conventions used.

- 3. A rectangular block 250 mm X 100 mm X 80 mm is subjected to axial loads as follows
 - 480 kN tensile in the direction of its length
 - 1000 kN compressive on the 250 mm X 100 mm faces and
 - 900 kN tensile on 250 mm X 80 mm faces.

Assuming the poisson's ratio as 0.25, find in terms of modulus of elasticity of the material E , the strains in the directions of each force.

If $E = 200 \text{ kN/mm}^2$, find the values of the modulus of rigidity and bulk modulus of the material of the block. Also, calculate the change in volume of the block due to loading specified above. *Prove the formulae used.*

- 4. A beam ABCD, 10 m long is supported at B and C. The overhangs AB and CD are 2m and 3m respectively. The overhang AB carries a U.D.L of 2 kN and CD carries a U.D.L of 1.5 kN. In addition, there are point loads of 1 kN, 2kN and 1 kN at distances of 1.5 m, 3m and 8 m from A respectively. Draw the shear force and bending moment diagrams and calculate the position and magnitude of maximum bending moment. Also locate the points of contraflexure, if any.
- 5. A cylindrical made of bronze 180 mm outside diameter and 15 mm thick is strengthened by a single layer of steel wire 2.25 mm diameter wound over it under a constant stress of 75 MN/m^2 . The Cylinder is subjected to an internal pressure of 27 MN/m^2 , with rise in temperature of the cylinder by 120°C . Assuming the cylinder to be a thin shell with closed ends, determine the final values of:
 - a. The stress in the wire
 - b. The circumferential stress in the cylinder wall
 - c. The radial pressure between the wire and the cylinder

$$E_s = 208 \text{ GN/m}^2; \alpha_s = 11.182 \times 10^{-6} \text{ per } ^\circ\text{C}$$

$$E_b = 90 \text{ GN/m}^2; \alpha_b = 18.6 \times 10^{-6} \text{ per } ^\circ\text{C}; \text{Poisson's ratio} = 0.32$$

- 6. Calculate the dimensions of the strongest section that can be cut out of a circular log of wood 25 cm in diameter.

Also specify the safe maximum span for the beam of rectangular section when it is to carry a U.D.L of 2.5 kN per metre and the bending stresses are limited to 10 MN/m^2 .

What will be the longest span if the circular log itself is used as a beam.

- 7. At a point in stressed material the direct stresses on two perpendicular planes are 140 MN/m^2 compression and 90 MN/m^2 tension respectively, and the shearing stress on these planes is τ . The yield stress for the material is 750 MN/m^2 . Find the value of τ at which failure may be expected according to different theories of failure. Take factor of safety = 4 and Poisson's ratio as 0.32. *Also give the graphical representation of the theories of failure.*
- 8. A cantilever of a circular section tapers uniformly from a diameter d at the free end to $2d$ at the fixed end. It carries a concentrated load W at the free end. Calculate the diameter of a cantilever uniform cross section, which would have the same deflection.